

ASSIGNMENT QUESTION

Easy Level

1. What is the purpose of a pivot table in excel?

Ans- A pivot table in Microsoft excel is used to **summarize data, organize , and analyze large sets of data quickly and interactively.**

- **Main purpose of a pivot table :**

➤ Summarize data

Convert long details tables into concise summaries .

Example- total sales by regions , product or salesperson .

Group and categories data

Organize data into categories (months, departments).

➤ Perform calculation

Automatically calculates sums , averages , counts , percentages , etc., without writing formula .

➤ Analyze patterns and trends

Identify relationships or insights (e. g. , which product sells best in each month).

➤ Reorganize data dynamically

Drag and drop fields to change how the data is displayed – rows , columns , values.

➤ Create reports

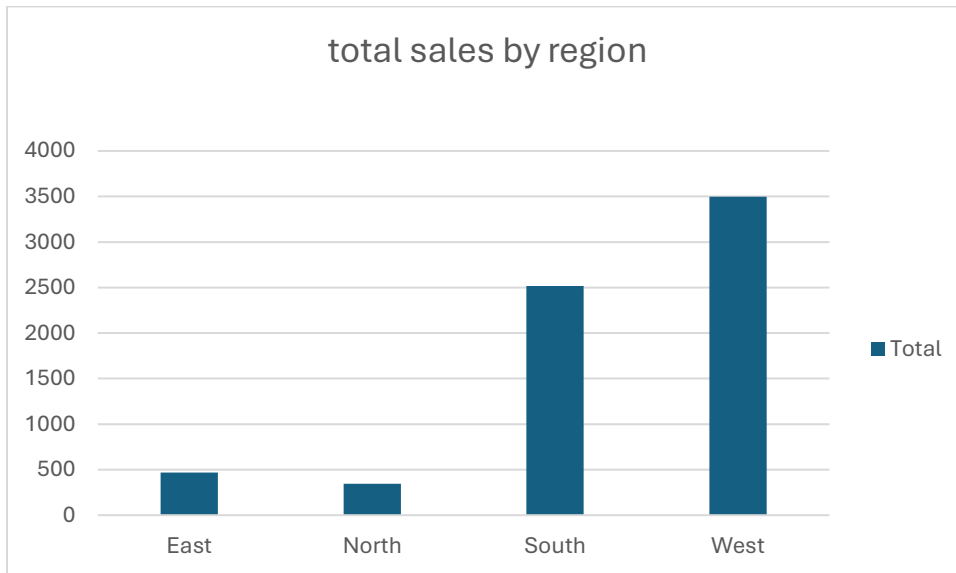
Turn row data into clean , readable summaries for decision-making.

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2. Create a pivot Table to show total sales by region?

Ans-

Row Labels	Sum of Sales
East	467
North	347
South	2518
West	3497
Grand Total	6829



3. Apply conditional formatting to highlight sales greater than 400.

Ans- **conditional formatting of sales greater than 400**

Date	Region	Product	Sales	Quantity
01-01-2024 00:00	West	Mangoes	456	93
10-01-2024 12:37	South	Oranges	348	56
20-01-2024 01:15	South	Mangoes	247	76
29-01-2024 13:53	West	Apples	230	37
08-02-2024 02:31	West	Mangoes	281	50
17-02-2024 15:09	West	Oranges	334	52

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27-02-2024 03:47	South	Apples	417	29
07-03-2024 16:25	West	Apples	441	44
17-03-2024 05:03	South	Mangoes	76	63
26-03-2024 17:41	West	Oranges	441	79
05-04-2024 06:18	South	Mangoes	205	23
14-04-2024 18:56	West	Oranges	483	97
24-04-2024 07:34	East	Mangoes	467	97
03-05-2024 20:12	West	Mangoes	380	92
13-05-2024 08:50	South	Oranges	497	71
22-05-2024 21:28	South	Mangoes	409	81
01-06-2024 10:06	West	Oranges	53	45
10-06-2024 22:44	West	Apples	398	33
20-06-2024 11:22	North	Oranges	347	35
30-06-2024 00:00	South	Apples	319	45

4. Explain the difference between sorting and filtering with an example from this dataset.

Ans-

✓ **sorting -**

➤ **definition-**

sorting is the process of arranging data in a specific order.

➤ **Purpose -**

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To organize data , making it easier to read or find information.

➤ **Effect on data –**

All rows remain visible , only the order changes .

Similar values come together , makes comparison easier.

➤ **Types of sorting-**

- **Ascending - (A-Z, smallest – largest)**
- **Descending – (Z-A, largest- smallest)**

➤ **Example**

❖ **A sample dataset for sorting –**

Date	Region	Product	Sales	Quantity
01-01-2024 00:00	West	Mangoes	456	93
10-01-2024 12:37	South	Oranges	348	56
20-01-2024 01:15	South	Mangoes	247	76
29-01-2024 13:53	West	Apples	230	37
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10-06-2024 22:44	West	Apples	398	33
20-06-2024 11:22	North	Oranges	347	35
30-06-2024 00:00	South	Apples	319	45

❖ After sorting the dataset by sales in descending order(high – low)

Date	Region	Product	Sales	Quantity
13-05-2024 08:50	South	Oranges	497	71
14-04-2024 18:56	West	Oranges	483	97
24-04-2024 07:34	East	Mangoes	467	97
01-01-2024 00:00	West	Mangoes	456	93
26-03-2024 17:41	West	Oranges	441	79
07-03-2024 16:25	West	Apples	441	44
27-02-2024 03:47	South	Apples	417	29
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29-01-2024 13:53	West	Apples	230	37
05-04-2024 06:18	South	Mangoes	205	23
17-03-2024 05:03	South	Mangoes	76	63
01-06-2024 10:06	West	Oranges	53	45

✓ Filtering

➤ Definition –

Filtering is a feature used to display only the specific data that matches given conditions while hiding the remaining data temporarily.

➤ Purpose of filtering

- Helps find required information quickly.
- Makes large data easier to analyze.

➤ Effects on data –

- Shows only selected records, hides unwanted data temporarily.
- Does not delete any data .
- Helps in faster analysis and reporting

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➤ Example

- A sample dataset of filtering

Date	Region	Product	Sales	Quantity
13-05-2024 08:50	South	Oranges	497	71
14-04-2024 18:56	West	Oranges	483	97
24-04-2024 07:34	East	Mangoes	467	97
01-01-2024 00:00	West	Mangoes	456	93
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05-04-2024 06:18	South	Mangoes	205	23
17-03-2024 05:03	South	Mangoes	76	63

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01-06-2024 10:06	West	Oranges	53	45
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❖ After filtering the data –

- Suppose we want the sales of fruits of west region

Date	Region	Product	Sales	Quantity
01-01-2024 00:00	West	Mangoes	456	93
29-01-2024 13:53	West	Apples	230	37
08-02-2024 02:31	West	Mangoes	281	50
17-02-2024 15:09	West	Oranges	334	52
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03-05-2024 20:12	West	Mangoes	380	92
01-06-2024 10:06	West	Oranges	53	45
10-06-2024 22:44	West	Apples	398	33

- Suppose we want the sales of mangoes acc. To regions :

Date	Region	Product	Sales	Quantity
01-01-2024 00:00	West	Mangoes	456	93
20-01-2024 01:15	South	Mangoes	247	76
08-02-2024 02:31	West	Mangoes	281	50
17-03-2024 05:03	South	Mangoes	76	63

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05-04-2024 06:18	South	Mangoes	205	23
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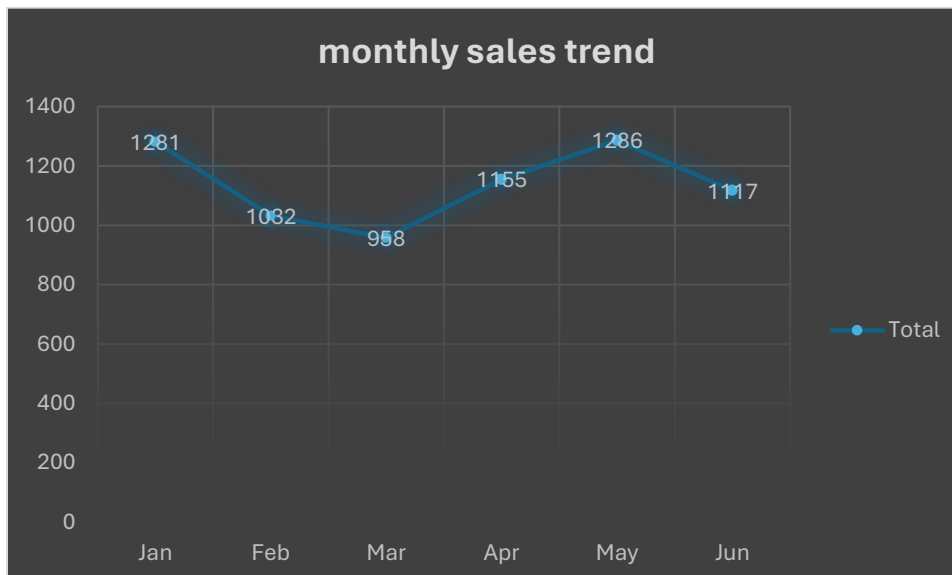
5 . create a pivot chart showing monthly sales trend.

Ans-

○

❖ Row Labels	Sum of Sales
Jan	1281
Feb	1032
Mar	958
Apr	1155
May	1286
Jun	1117
Grand Total	6829

○ Pivot chart : line chart to find sales trend



- What this pivot chart helps you see

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- ✓ Sales growth or decline over months .
- ✓ Seasonal trends
- ✓ High and low performing months .

6. Use a slicer to filter data by product and observe changes in sales .

Ans-

❖ Pivot table

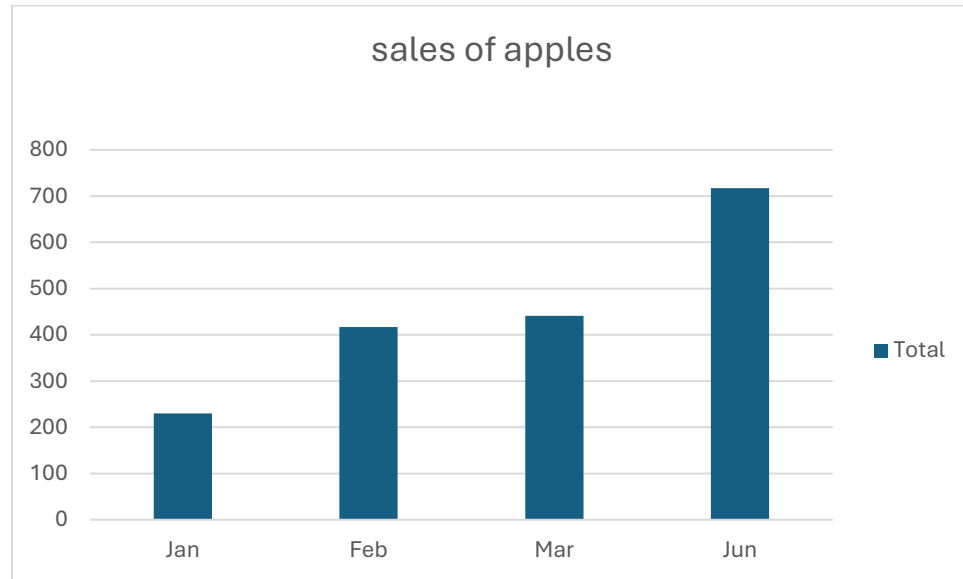
Row Labels	Sum of Sales
Jan	230
Feb	417
Mar	441
Jun	717
Grand Total	1805

❖ Slicers



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❖ Line chart to show the sales of apples



7. Why are dashboards important for decision-making in business analysis?

- **Ans- Dashboards are important in business analysis because they provide a clear , visual , and real-time summary of key business information helping stakeholders make informed , faster , and more accurate decision .**
- **Quick overview of business performance –**
A dashboard shows key data like sales , profit , expenses , or customer numbers in one place. Manager can understand what's happening quickly .
- **Better decision- making –**
With real-time or update data , businesses can make faster and smarter decisions instead of guessing .
- **Easy to find trends and patterns –**
Dashboards use charts / graphs , so it's easier to spot increasing sales, falling profits, seasonal trends ,etc.

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- **Tracks important goals (KPIs)-**
Business can monitor targets like monthly sales , employee performance , or customer satisfaction .
- **Saves time –**
Instead of checking many excel sheets or reports , everything is available in one dashboards.
- **Helps solve problems early –**
If sales suddenly drop or costs rise , the dashboards highlights it quickly so action can be taken .

❖ **Summary table**

benefits	explanations
quickly insights	data becomes easies to read
real- time monitoring	react faster
better decision	informed choices
track KPIs	monitor goals
identify trends	spot patterns
saves time	less manual works
effective communication	clear data visualization
interactive	drill down & filter

8 . create a dashboard that includes :

- Sales by region (pivotable + charts)
- Product filter (slicer)
- Monthly trend (line chart)

Ans- [excel.xlsx](#)

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9. Group dates in the pivot table into months and calculate the average sales per month ?

Ans-

Row Labels	Average of Sales
Jan	320.3
Feb	344.0
Mar	319.3
Apr	385.0
May	428.7
Jun	279.3
Grand Total	341.5

10. combine apples and oranges into one group called fruits and compare its sales with other products .

Ans-

Row Labels	Sum of Sales
fruits	4308
Mangoes	2521
	6829

